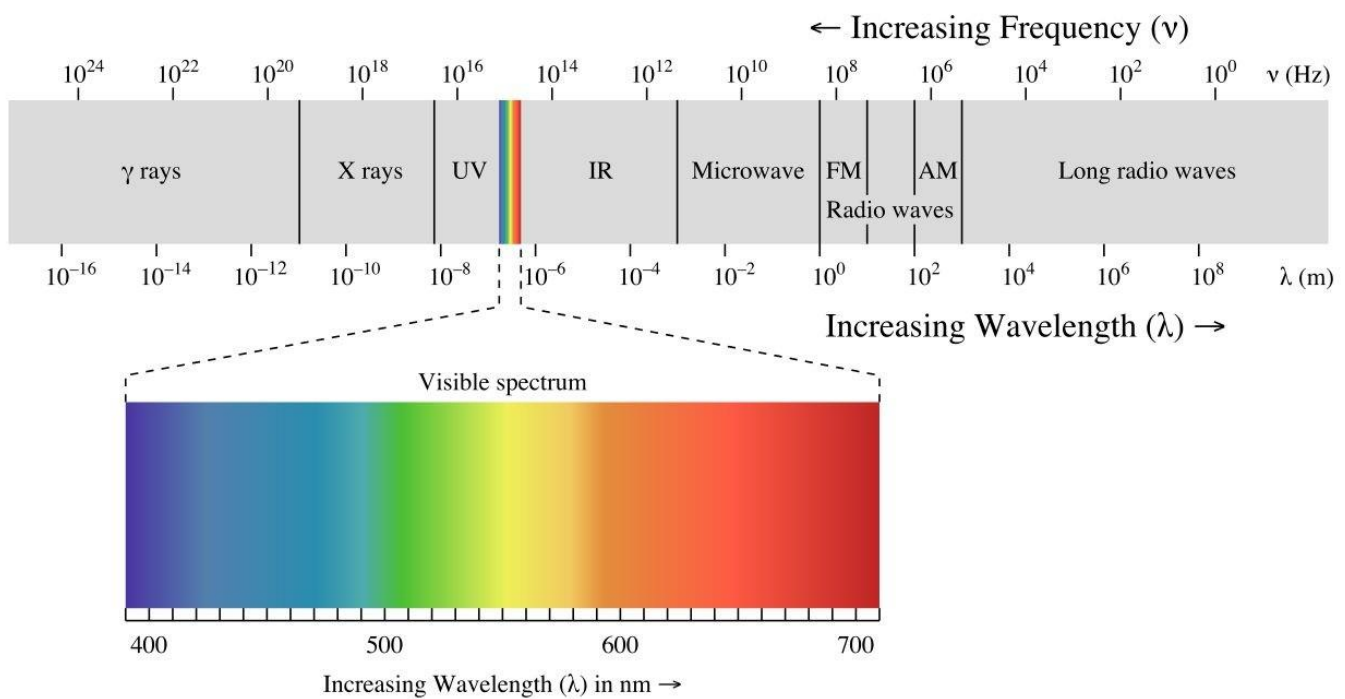


Unit 11

"The modern physics"



Lesson 1 “black body radiation phenomena”

Groub 1:

write the scientific term expressed by the following statements

no	Statement	answer
1	It is the branch of physics that explains everything in our daily life and our common experiences	Classical physics
2	It is the branch of physics that deals with scientific phenomena in the universe which might not be directly observed in our daily life, and classical physics cannot explain	Modern (quantum) physics
3	They are vibrating electric and magnetic fields that move through space or any other medium	Electromagnetic spectrum
4	The wavelength at maximum radiation intensity ($\lambda_{\max}$) is inversely proportional to the temperature	Wien's Law
5	The wavelength ($\lambda_{\max}$) at which the peak of the curve occurs is inversely proportional to temperature	Wien's Law
6	It is an ideal system that absorbs all radiation (100% of the radiation that hits it) falling on it. Then re-emit this radiation totally again	Black body radiation
7	The distribution of radiation intensity with wavelength	Planck's distribution
8	A body that absorbs all the radiation falling on it with different wavelengths emits them again in an ideal form.	Black body
9	The transfer of heat by electromagnetic waves such as visible light, infrared, and ultraviolet radiation.	Radiation
10	the ratio between the photon energy to its frequency.	Planck's constant (h)
11	A phenomenon used for criminology and night vision	Thermal imaging

Groub 2:

mention one factor that can increase

no	Statement	answer
1	Energy of the photon	Increasing frequency Or Decreasing wavelength
2	The color of the hot objects	Temperature
3	The intensity of the black body radiation	wavelength
4	The wavelength at maximum radiation intensity ($\lambda_{\max}$)	Decreasing temperature

Groub 3:

Choose the correct answer

- 1- Which of the following electromagnetic waves have the shortest wavelength
a) gamma ray b) X-Rays c) UV d) Radio
- 2- The electromagnetic radiation that have the highest frequency and the shortest wavelength.....
a) gamma ray b) X-Rays c) UV d) Radio
- 3- Due to the classical expectation the intensity of radiation increases by.....
a) Decreasing wavelength of radiation
b) Decreases frequency of radiation
c) Increases wavelength of radiation
- 4- the wavelength of a photon is 2.2×10^{-7} m, its energy is given by
a) 1×10^{-35} joule b) 9×10^{-19} joule c) 3.3×10^{-27} joule d) 4.4×10^{-32} joule
- 5- Difference between modern physics and quantum physics
a) Modern physics talks about macroscopical model
b) Quantum physics talks about macroscopical model
c) Both are the same
- 6- Difference between modern physics and classical physics
a) Modern physics talks about macroscopical model
b) Classical physics talks about macroscopical model
c) Both are the same
- 7- The emitted energy of the sun radiation is ----- visible light.
a) 40% b) 50% c) 20% d) 80%
- 8- The emitted energy of the sun radiation is ----- heat radiation
a) 40% b) 50% c) 20% d) 80%
- 9- The emitted energy of the lamp radiation is ----- visible light
a) 40% b) 50% c) 20% d) 80%
- 10- The emitted energy of the lamp radiation is ----- heat radiation
a) 40% b) 50% c) 20% d) 80%
- 11- The electromagnetic radiation that have the lowest frequency and the tallest wavelength.....
a) gamma ray b) X-Rays c) UV d) Radio

12- if the body is ideal black body, so its dominant color for the radiation that radiate from it is changes according to

- a) the mass of the body b) The density of the body
c) The temperature of the body d) The type of material that makes the body

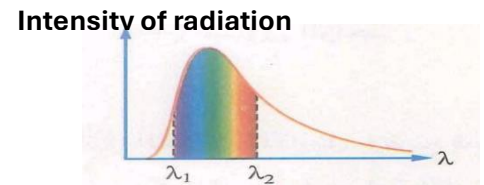
13-In the plank's curve for the black body radiation, if the temperature decreased

	The peak of the curve	The space under the curve
a	Displaced to high frequencies region	Increases
b	Displaced to high frequencies region	Decreases
c	Displaced to Low frequencies region	Increases
d	Displaced to Low frequencies region	Decreases

14- In the Planck's curve for the black body radiation, the intensity of radiation at the very high frequencies is

- a) Doesn't change b) Moderate. c) Very high d) Close to zero.

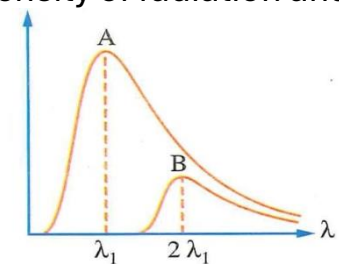
15- At the opposite graph, if λ_1 is the shortest wavelength of the visible rays, while λ_2 is the longest wavelength for visible rays, so the graph might represent the radiation that radiate from



- a) A glowing star b) Earth c) Tungsten lamp d) Human body.

16- The opposite graph represents the relation between the intensity of radiation and the wavelength for two hot bodies A, B,

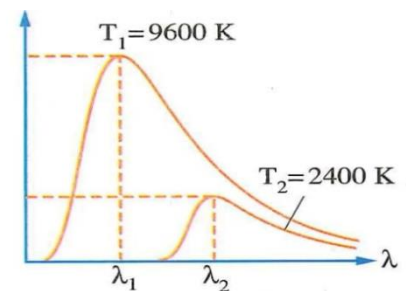
so the ratio between their temperatures ($\frac{T_A}{T_B}$) is



- a) $\frac{4}{1}$ b) $\frac{2}{1}$ c) $\frac{1}{2}$ d) $\frac{1}{4}$

17- The opposite graph shows a hot black body at T_1, T_2

so the ratio between ($\frac{\lambda_1}{\lambda_2}$) is



- a) $\frac{4}{1}$ b) $\frac{2}{1}$ c) $\frac{1}{2}$ d) $\frac{1}{4}$

18- A black body radiation with 3000 K, and its maximum wavelength λ , if the body is cooled to be T, its maximum wavelength become 10λ , so T equals

- a) 300 K b) 2700 K c) 270 K d) 1800 K

19-The frequency of photons which emitted from a glowing body increase, then its number

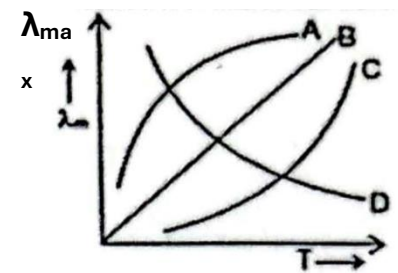
- a) Increases b) Remains constant c) Decreases d) Equals zero

20- Which of the following curves represents the classical explanation for blackbody radiation, and which one is the modern explanation

- a) A: modern B: classical b) A: classical B: classical
c) A: modern B: modern. d) A: classical B: modern

21- For a black body radiation if $\lambda_{\max}$ represents the maximum wavelength corresponding to maximum radiation at temperature T, the curve between $\lambda_{\max}$ and T is

- a) A
b) B
c) C
d) D



Groub 4:

True and false

- 1-All objects radiate energy continuously in the form of electromagnetic waves produced by thermal vibrations of the molecules ()
- 2-Wave particle duality states that a wave can only act as a wave and not a particle. ()
- 3-The electromagnetic waves are transverse and longitudinal waves ()
- 4-The electromagnetic waves need medium to travel through it. ()
- 5-X-rays is faster than radio ()
- 6-White surfaces reflect all colors ()
- 7-When any radiation is absorbed, its energy is changed into heat energy. ()
- 8-Black body is a Prefect absorber and prefect emitter ()
- 9-Electromagnetic source emit all wavelengths equally ()
- 10-At low temperatures, the wavelengths of the thermal radiation are mainly in the visible light region ()

- 11- The intensity of radiation is related to the wavelength ()
- 12- At short wavelengths, classical theory is in good agreement with the experimental data. ()
- 13- the energy of electromagnetic waves is quantized ()
- 14-In Wien's law, the smaller temperature, the smaller the wave length of the peak ()
- 15-The atom does not radiate as long as it remains in one energy level ()

Groub 5:

Solve the following

- 1- If the temperature of surface of sun is 6000 K and wavelength ($\lambda_{\max}$) at maximum radiation intensity is 0.499 nm. Calculate the temperature of filament lamp that gives photons at wavelength ($\lambda_{\max}$) = 1.2 nm
.....
.....
- 2- if the wavelength having a maximum intensity in the solar radiation is 0.5 μm Calculate the wavelength having a maximum intensity in the radiation emitted by Earth, given that the surface temperature of the sun and Earth are 6000 K and 300 K respectively
.....
.....
- 3- calculate the wavelength of photon with energy = 4.968×10^{-19} Joule
.....
.....
- 4- A body at temperature 1500 K radiates out maximum energy at the wavelength 20000 Å. If the sun radiates out maximum energy at 5000 Å, calculate the temperature of the sun
.....
.....
- 5- What is the energy of a photon with frequency $3.0 \times 10^{14} \text{ Hz}$
.....
.....
- 6- The wavelength of yellow light is approximately 5.9×10^{-7} meters. What is its frequency
.....
.....

Lesson 2 “thermionic emission”

Groub 1:

write the scientific term expressed by the following statements

no	Statement	answer
1	emission of electrons from a heated metal's surface (by heating)	Thermionic emission
2	emission of electrons from a metallic surface when the light energy falling on it	photoelectric effect
3	It is the attractive force between the positive ions and free electrons in metal which make the electrons cannot leave it	Surface potential barrier

Groub 2:

mention one factor that can increase

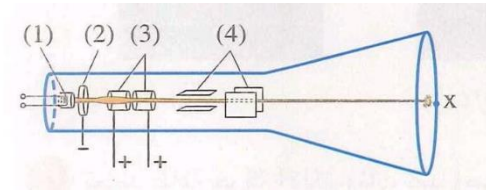
no	Statement	answer
1	The intensity of this illumination (illumination of the fluorescent screen)	kinetic energy of the incident electron
2	the kinetic energy of the electron in CRT	the potential difference between the cathode and the anode

Groub 3:

Choose the correct answer

- The cathode rays consist of
 - Electrons
 - photons
 - Gamma
 - electromagnetic wave
- The process in which electrons are emitted from a metallic surface when heat given to it is called
 - photoelectric effect
 - thermionic effect
 - Compton effect
 - Black body radiation
- CRT stands for
 - photoelectric effect
 - thermionic effect
 - Compton effect
 - Black body radiation
- In the C.R.T when the filament is damaged
 - Intensity of illumination increases on fluorescent screen
 - Intensity of illumination decreases on fluorescent screen
 - The fluorescent screen doesn't illuminate
 - The deflection of electron beam decreases

5-The opposite figure represents C.R.T, which of the following parts represents.....



A. The source of electrons.....

- a) part 1 b) part 2 c) part 3 d) part 4

B. The part that responsible for changing the position of collision of electron beam with fluorescent screen.....

- a) part 1 b) part 2 c) part 3 d) part 4

C. The illumination intensity is controlled by At point X.

- a) part 1 b) part 2 c) part 3 d) part 4

6- In C.R.T the velocity of electron is v , when the electron is accelerated by potential difference with V , if the potential difference is increased to $2V$, son the velocity will become.....

- a) $2V$ b) $\sqrt{4v}$ c) $4V$ d) $\frac{1}{2}V$

7- If the potential difference between anode and cathode in C.R.T is 1000 V : so...,

A. K.E for the electron is

- a) 1.6×10^{-15} b) 1.6×10^{-16} c) 1.6×10^{-17} d) 32×10^{-16}

B. The maximum velocity of liberated electron from cathode when it reaches anode equals.....

- a) 1.88×10^6 b) 1.88×10^7 c) 3.52×10^5 d) 3.52×10^4

8- The source of thermal energy in the C.R.T is.....

- a) Anode b) Cathode c) Filament d) Fluorescent screen

9- It is used to accelerate the electron beam towards the screen.....

- a) Anode b) Cathode c) Filament d) Fluorescent screen

10- To increase to velocity of electron in C.R.T for four times.....

- a) Increase K.E for four times b) increase of 16 times
c) Increase the potential difference to double d) Increase K.E for to double

Groub 4:

True and false

- 1- Filament It is used to heat up the cathode ()
- 2- Anode Control the number of electrons in the electron beam ()
- 3- filment It is used to accelerate the electron beam towards the screen circuit ()
- 4- Electron gun It is an instrument to produce a beam of electrons at high speed ()
- 5- Fluorescent Screen Emits green light when the electron strikes it ()

Groub 5:

Solve the following

- 1- An electron moving at a constant speed of 4.4×10^5 m/sec, what is its kinetic energy?

.....

- 2- Calculate the speed of an electron that has kinetic energy of 120 eV

.....

- 3- calculate the potential difference which accelerated an electron from rest with speed equals 9.796×10^6 m/s

.....

Groub 6:

complete the following sentences

- 1- There are many ways to eject (escape) electrons out of a material
, and
- 2- is application on the thermionic emission.
- 3- C.R.T is used to make and Monitors
- 4- is used to control the number of electrons of the electron beam.
- 5- is used to accelerate the electron beam towards the screen
- 6- The stage at which the beam at high speed.....
- 7-emits green light when the electron strike it

Lesson 3 “photoelectric effect”

Groub 1:

write the scientific term expressed by the following statements

no	Statement	answer
1	emission of electrons from a heated metal's surface (by heating)	Thermionic emission
2	emission of electrons from a metallic surface when the light energy falling on it	photoelectric effect
3	An electron released or ejected from a substance by photoelectric effect	Photoelectrons
	It is the minimum energy needed to free (release) the electrons from the surface of the metal	work function of the metal

Groub 2:

mention one factor that can increase

no	Statement	answer
1	The kinetic energy of the emitted electrons	the frequency of the incident light and the work function of the metal
2	the number of photoelectrons emitted per second	directly proportional to the intensity of the radiation

Groub 3:

Choose the correct answer

1- The process in which electrons are emitted from a metallic surface when a light given to it is called.....

- a) photoelectric effect b) thermionic effect c) Compton effect d) Black body radiation

2- the emission of electrons from metallic surface (photoelectric effect) depends on

- a) The intensity of the incident light b) The frequency of incident light
c) The energy of incident light d) B&C

3- An electromagnetic radiation is incident on a photoelectric cell with ν , photoelectrons emitted with $K.E_{Max}$, if the frequency becomes 2ν , which of the following doesn't change

- a) The photon energy b) Speed of photon
c) $K.E_{Max}$ for emitted photoelectron d) Maximum velocity of emitted photoelectron

3- A metal has a work function 8×10^{-19} J. Calculate the maximum kinetic energy of electrons emitted from the metal when light of frequency 2×10^{15} Hz is shone on the surface

.....

.....

4- When monochromatic light of wave length 663 nm is incident on a metal plate whose work function 3.52×10^{-19} j will electrons be emitted or not

.....

.....

Groub 6:

complete the following sentences

- 1- An application on photoelectric effect is
- 2- photoelectric cell aims to convert To
- 3- the source of free electrons in photoelectric cell is
- 4- the collector in photoelectric cell is called
- 5- At the case of the relationship between light intensity and photocurrent is independent of each other
- 6- At the case of the relationship between light intensity and photocurrent is directly proportional to each other

Groub 7:

Compare between the following

	$E > E_w$	$E < E_w$
The relationship between light intensity and photocurrent		

Lesson 4 “Compton effect”

Groub 1:

write the scientific term expressed by the following statements

no	Statement	answer
1	The total linear momentum is the same before and after collision	The conservation of momentum
2	The total energy is the same before and after collision	The conservation of energy

Groub 2:

Choose the correct answer

1- The ratio between the energy of the photon to the velocity of light in air is the of the photon

- a) mass b) kinetic energy c) frequency d) Momentum

2- in Compton effect, ratio between the photon energy after collision to its energy before collision is

- a) Greater than one b) Equal one c) less than one d) zero

3- In Compton effect, the ratio between the velocity of photon after collision to the velocity of photon before collision

- a) Greater than one b) Equal one c) less than one d) zero

5-In Compton effect, the ratio between the frequency of photon after collision to the frequency of photon before collision

- a) Greater than one b) Equal one c) less than one d) zero

6- In Compton effect, the ratio between the wavelength of photon after collision to the wavelength of photon before collision

- a) Greater than one b) Equal one c) less than one d) zero

7- in Compton effect, the speed of gamma rays photons after collides with a free electron is

- a) increases b) decreases c) doesn't change d) Equal zero

8-The Compton effect proves

- a) The particle nature of photon b) The wave nature of photon
c) The wave nature of electron d) The particle nature of electron

9- In Compton effect, when a photon with high frequency collides with free electron, which of the following will increase for photon after collision

- a) Energy b) wavelength c) frequency d) Momentum

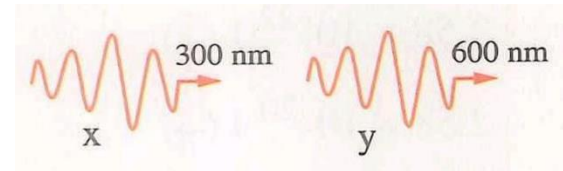
10- In Compton effect after collision, there is no decrease of

- a) The wavelength of electron b) Energy of photon
c) Photon frequency d) Photon speed

11- From the properties of photon

- a) Its speed equals the speed of light b) Deflects by the electric field
c) It may be accelerated d) All the above

12- At the opposite figure, the ratio between the speed of photon x and y is.....



- a) $\frac{1}{2}$ b) $\frac{2}{1}$ c) $\frac{1}{4}$ d) $\frac{1}{1}$

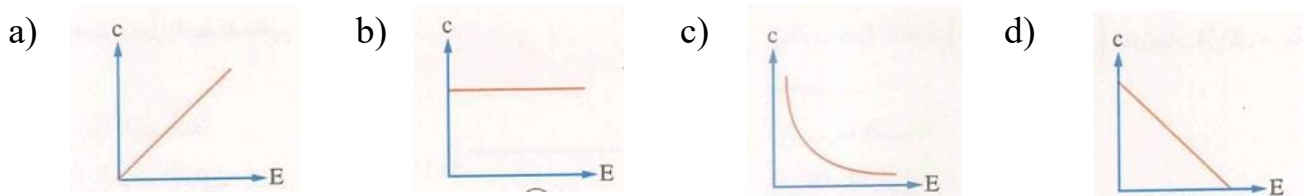
13- The linear momentum for a photon is

- a) $\frac{h}{\lambda}$ b) $\frac{h\nu}{\lambda}$ c) $\frac{hc}{\lambda}$ d) $\frac{h\nu}{c^2}$

14- Two photon the ratio between their frequencies is 1:2, so the ratio between their energies is

- a) 1: 2 b) 2: 1 c) 1: 1 d) 4: 1

15- Which of the following graphs represents the relation between the speed of photon and its energy



16- Which of the following graphs represents the relation between the energy of photon and its wavelength.....



Groub 3:

True and false

- 1- When a photon of high frequency (x- rays or gamma rays) collides with a free electron, The photon's frequency decreases ()
- 2- When a photon of high frequency (x- rays or gamma rays) collides with a free electron, The photon's wavelength decreases ()
- 3- When a photon of high frequency (x- rays or gamma rays) collides with a free electron, the photon's speed decreases . ()
- 4- When a photon of high frequency (x- rays or gamma rays) collides with a free electron, the electron's speed increases ()
- 5- Classical theories can explain the change in wavelength. ()

Groub 4:

Solve the following

- 1- calculate the photon energy and linear momentum if $\lambda = 380 \text{ nm}$

Groub 7:

Compare between the following

In Compton effect *after collision*

	Speed of electron	Speed of photon
The change		